
E-commerce Shipping via Classification Analysis

MIS 451 - Machine Learning for Business

Lecturer: Ms. Huynh Gia Linh
Quarter 3: 2025-2026

Presented By :
Group Lazy





Team Members

Võ Thị Hoài Anh - 2132300280

Phạm Thúy Huyền - 2132309006

Lê Anh Khương - 2332309001

Table of Content

1. Business Context and Objectives
2. Data Collection and Characteristics
3. Exploratory Data Analysis (EDA)
4. Data Transformation
5. Model Development
6. Model Evaluation and Comparison
7. Interpretation and Business Insights



Business Context and Objectives



Business Context

1. Organizations rely on ML for data-driven decisions to cut costs & improve efficiency.
2. Project: build a classification model predicting delivery outcomes from historical data.
3. Dataset: 5,000+ observations — supports full ML workflow from prep to evaluation.



Project Objectives

1

Data Analysis

Explore variable structure, distributions, and relationships
→ identify key delay factors.

2

Model Development

Compare KNN, SVM, Logistic Regression + ANN
→ select best performing model

3

Business Recommendations

Convert findings into actionable steps for the operations team.

Dataset Description

Kaggle

E-Commerce Shipping Dataset

10,999

Total rows

12

variables

Variable	Type	Description
Warehouse_block	Categorical	Warehouse Area (A–F)
Mode_of_Shipment	Categorical	Ship / Flight / Road
Customer_care_calls	Numerical	Number of support calls
Customer_rating	Numerical	Rating score (1–5)
Cost_of_the_Product	Numerical	Product price (USD)
Prior_purchases	Numerical	Previous purchase count
Product_importance	Categorical	Low / Medium / High
Discount_offered	Numerical	Discount percentage (%)
Weight_in_gms	Numerical	Package weight (grams)
Reached.on.Time_Y.N ♦	Target (Binary)	0 = On-time 1 = Late

Table 1. Description of variables in the E-Commerce Shipping Dataset

Exploratory Data Analysis (EDA)



Dataset Overview

The dataset was thoroughly checked for structure and quality immediately after loading.

Split: Train 8,799 objects (80%) and Test 2,200 objects (20%), using target stratification.

Target distribution: Class 1 (Delayed) = 59.7% · Class 0 (On Time) = 40.3%

Reason for choosing F1-macro: Slight imbalance 60/40 → accuracy can be misleading.

Train-Test Split

Split is performed before all transformation steps → ensuring the test set is completely independent.

Parameters: test_size=0.2, stratify=target, random_state=99

Principle: Information from the test set (mean, distribution) should not affect the preprocessing train.

Data Cleaning

`df.duplicated().sum() = 0` → no duplicate rows. Loop

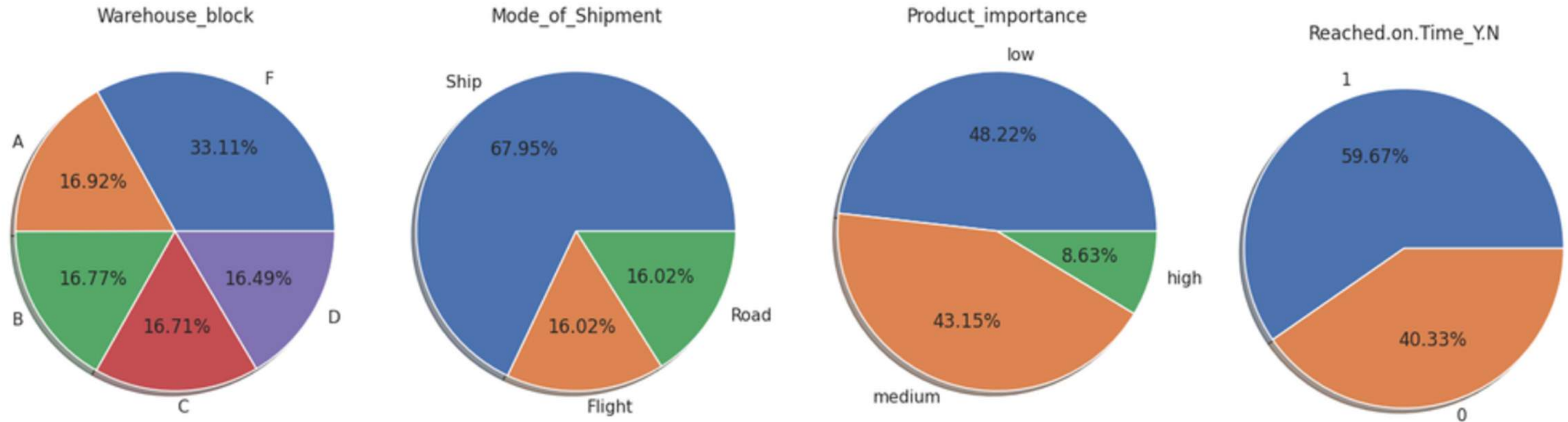
`snull().sum() = 0` across the entire column, both training and testing → no missing values

Result: Skip the imputation step, focus entirely on feature engineering and modeling

Exploratory Data Analysis (After cleaning)

Univariate Analysis

Categorical columns

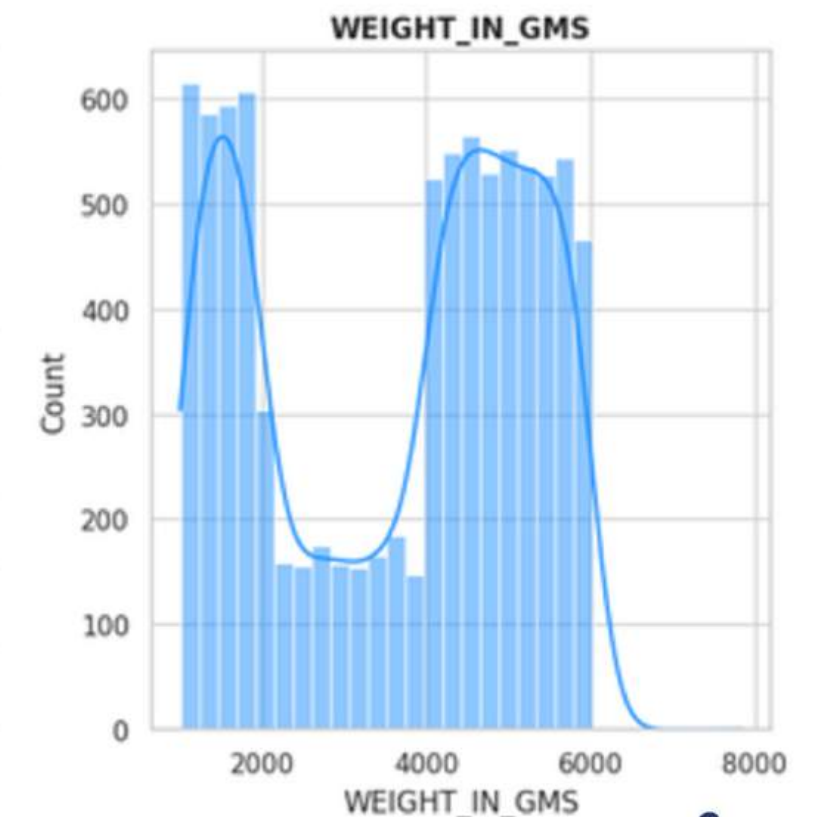
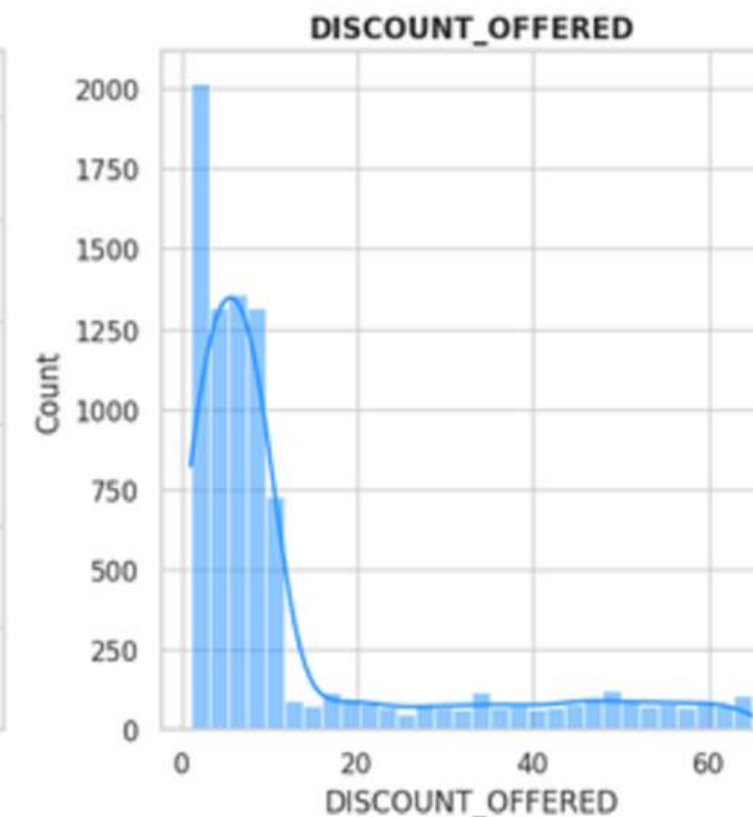
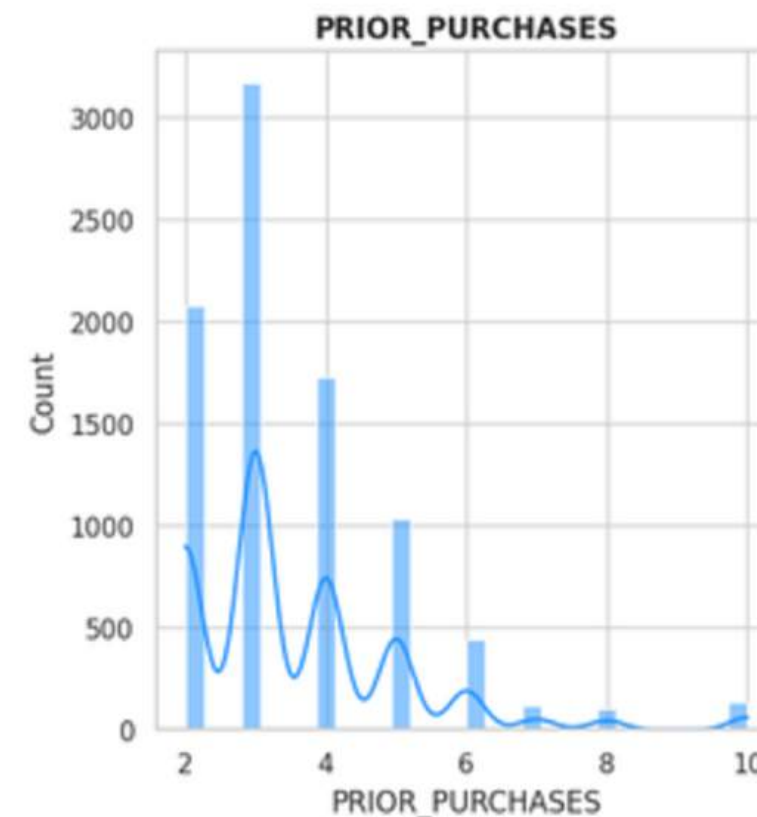
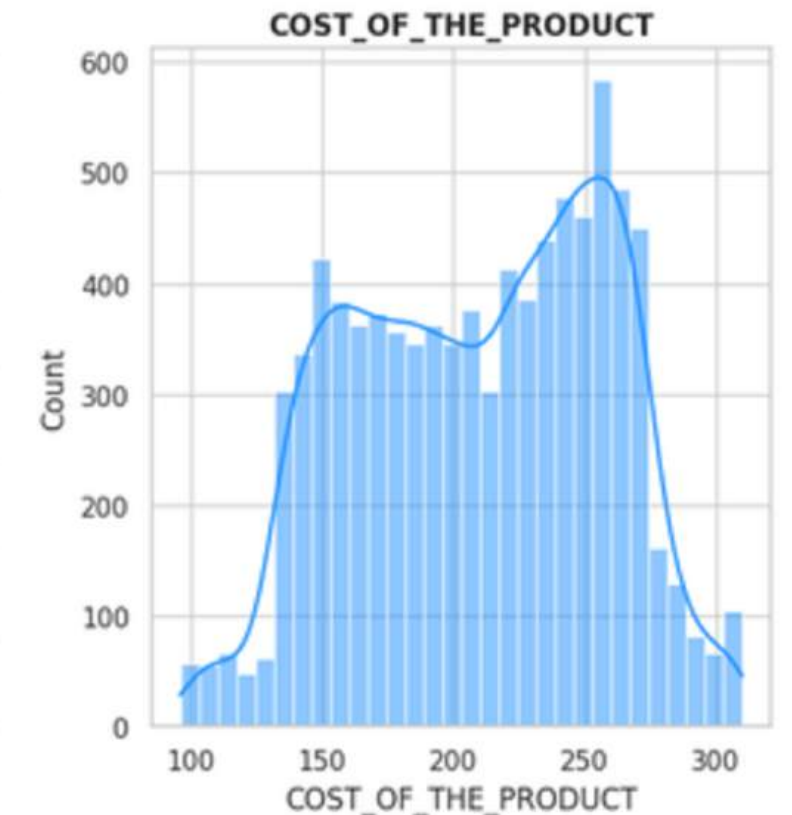
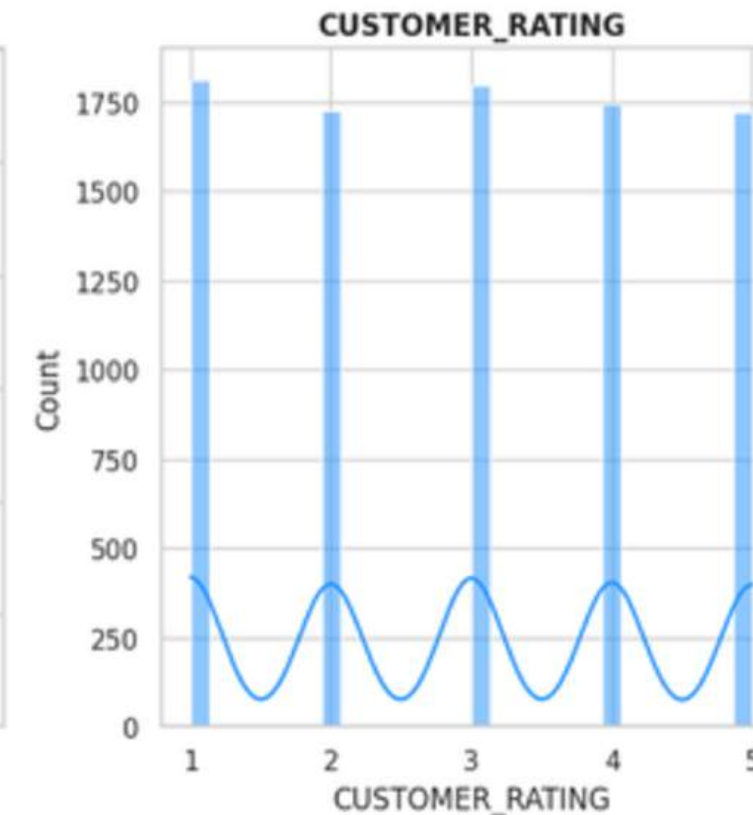
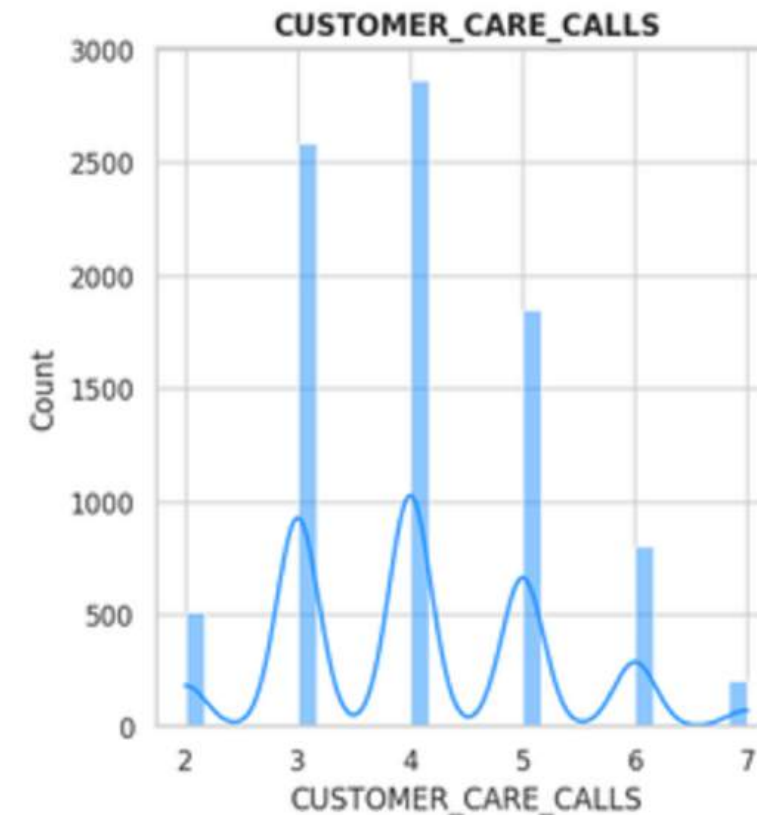


Current logistics are imbalanced, heavily relying on **Warehouse F (33%)** and **sea transport (68%)**, leading to a high risk of delays. For modeling, the features are well-represented, but the slight target imbalance (**~60% late deliveries**) necessitates careful evaluation metric selection.

Univariate Analysis

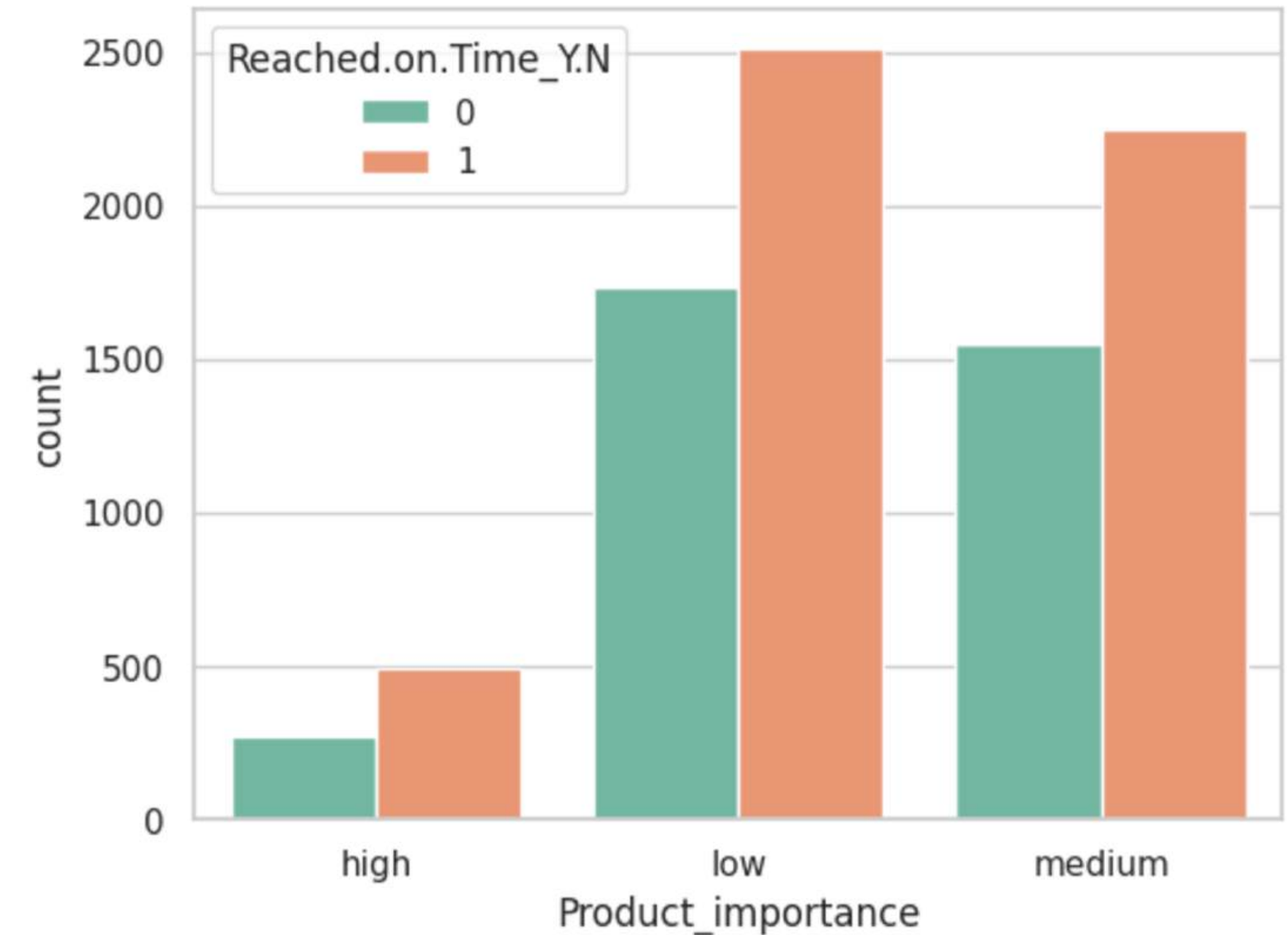
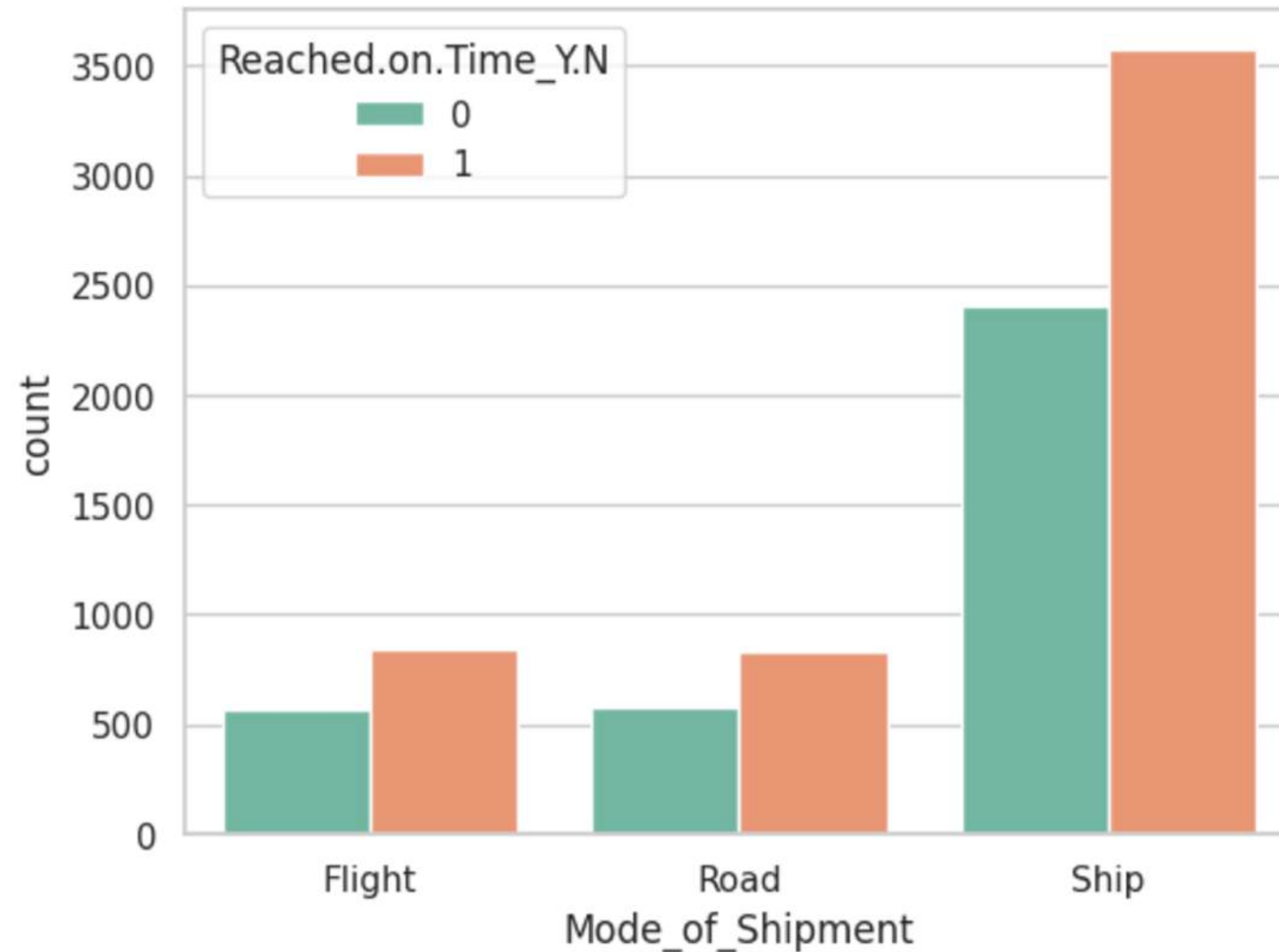
Numerical columns

- Discounts: Right-skewed; mostly 0–30%, with extreme outliers (>50%).
- Support Calls: High frequency (3–5 calls), signaling potential processing issues.
- Cost: Symmetrical (\$150–\$300), typical for mass-market goods.
- Weight: Fairly uniform spread (1,000g – 6,000g).



Bivariate Analysis

Categorical columns



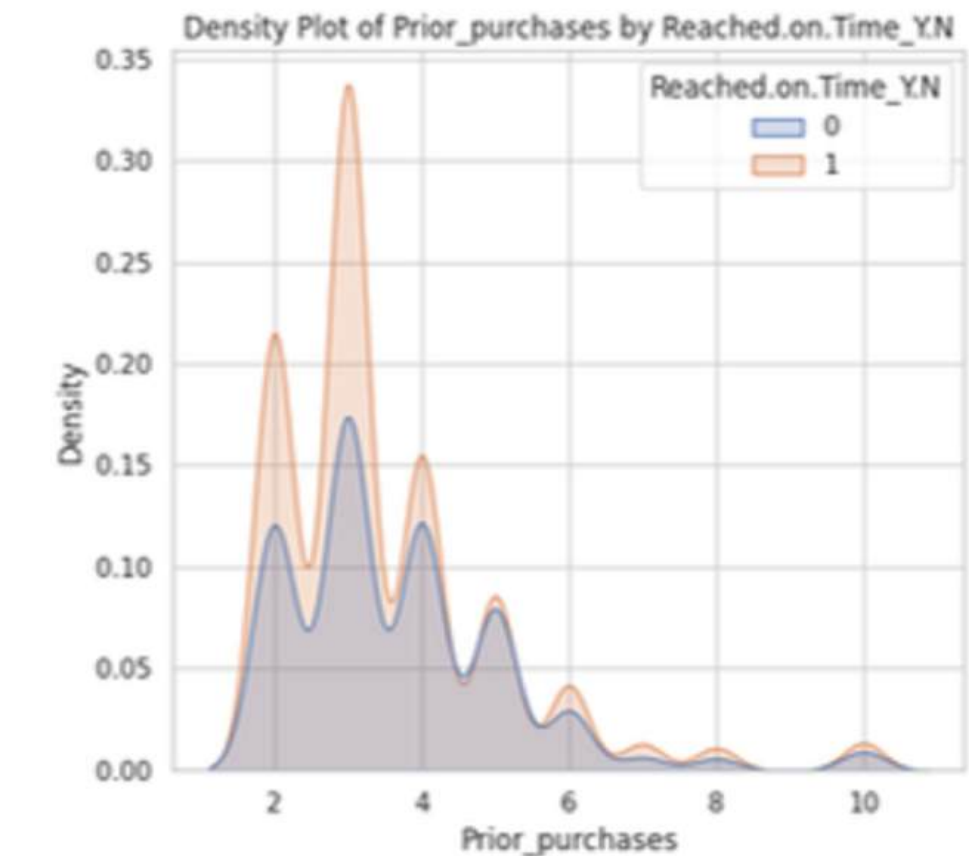
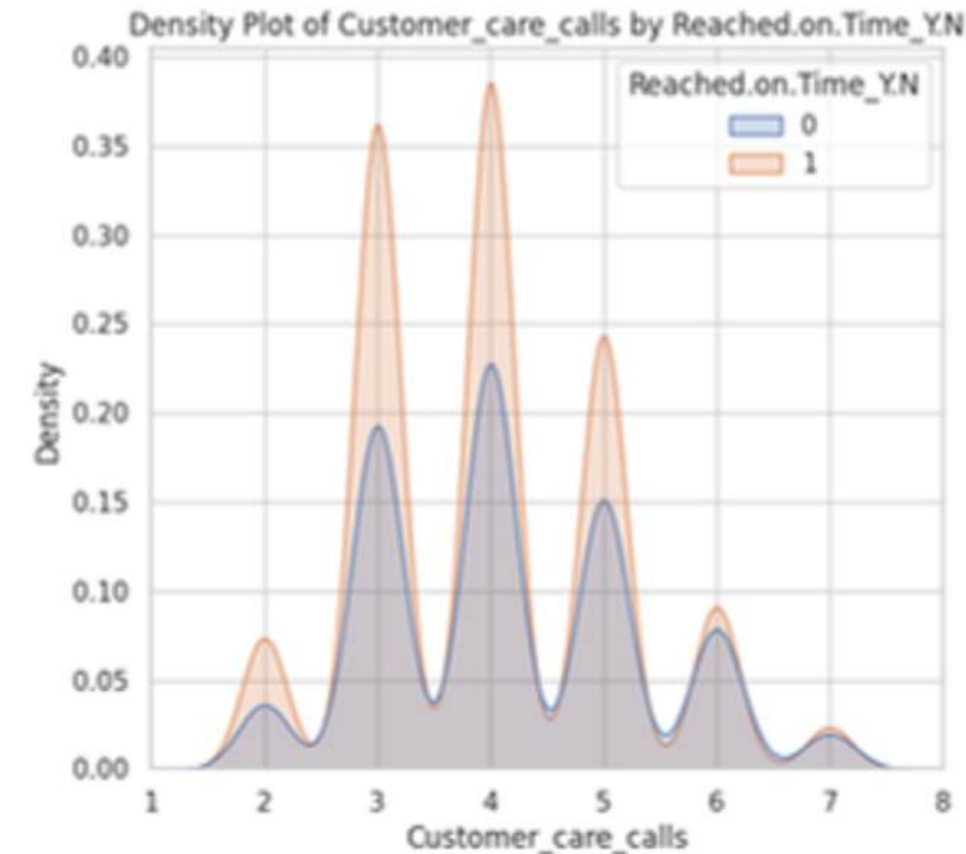
Delays consistently outnumber on-time deliveries across all shipping methods and importance levels. Because this ratio remains uniform, these features offer **low standalone predictive power**.

Bivariate Analysis

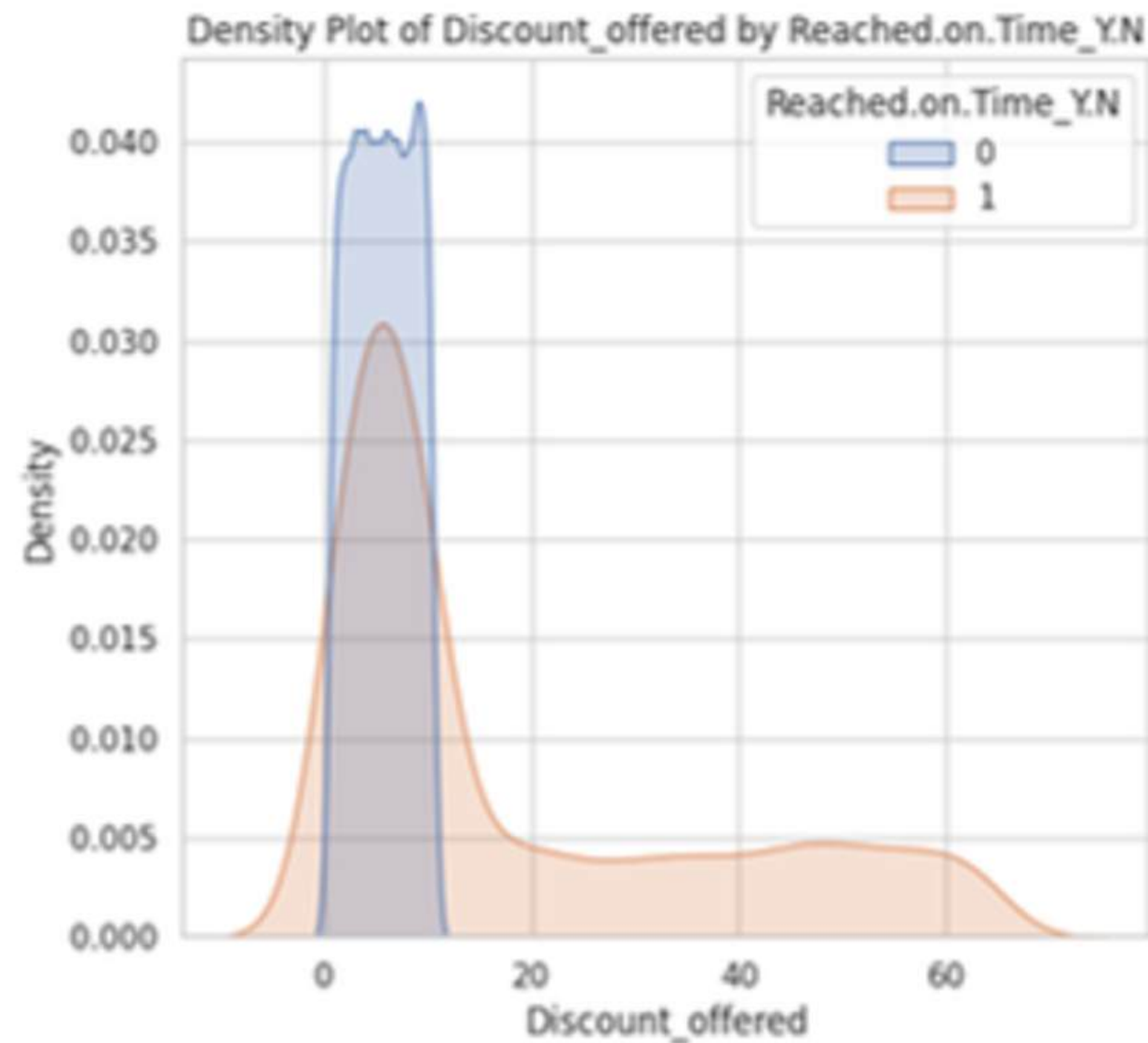
Numerical columns

Weak Predictors (High Overlap)

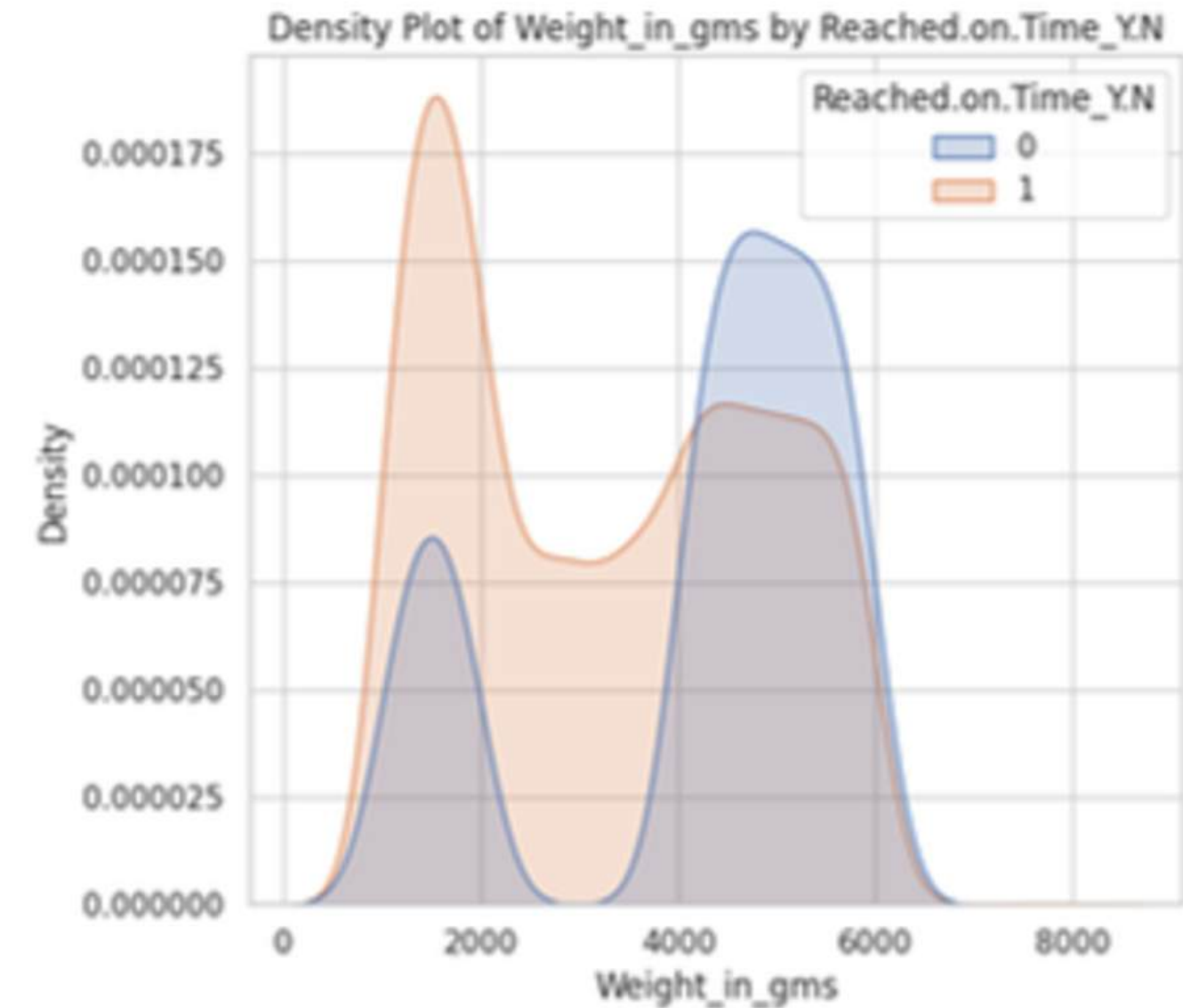
- Calls, Ratings & Prior_purchases: Distributions for on-time and late deliveries are almost identical, meaning they are not clear indicators of delays.
- Cost_of_the_Product: Despite minor fluctuations, the cost distributions for both delivery outcomes remain heavily intertwined.



Bivariate Analysis



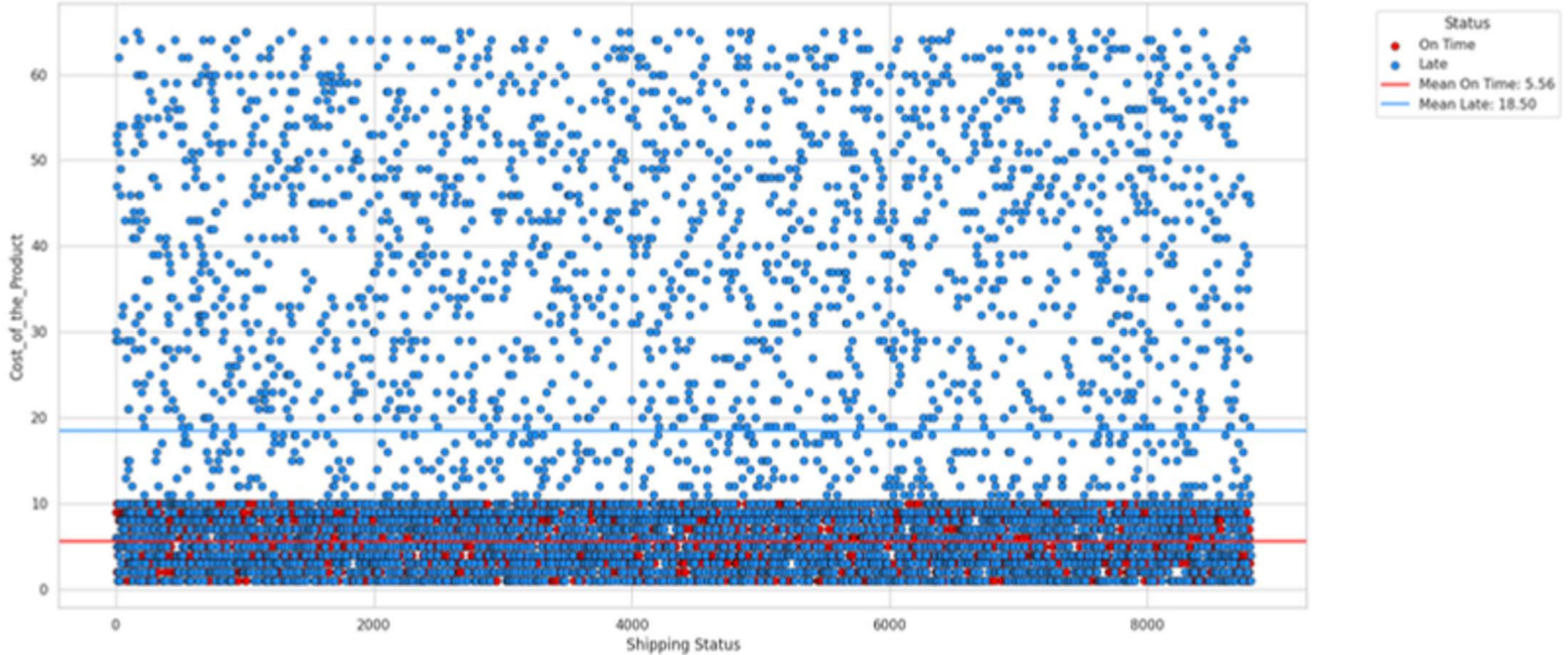
Discount_offered (**Most Important**): On-time deliveries peak sharply at 0–10%. Products with discounts >10% are almost always delivered late.



Weight_in_gms (**Bimodal**): Late deliveries cluster in lighter goods (1–3kg), while on-time deliveries dominate heavier goods (4–6kg). The 3–4kg range has very little data.

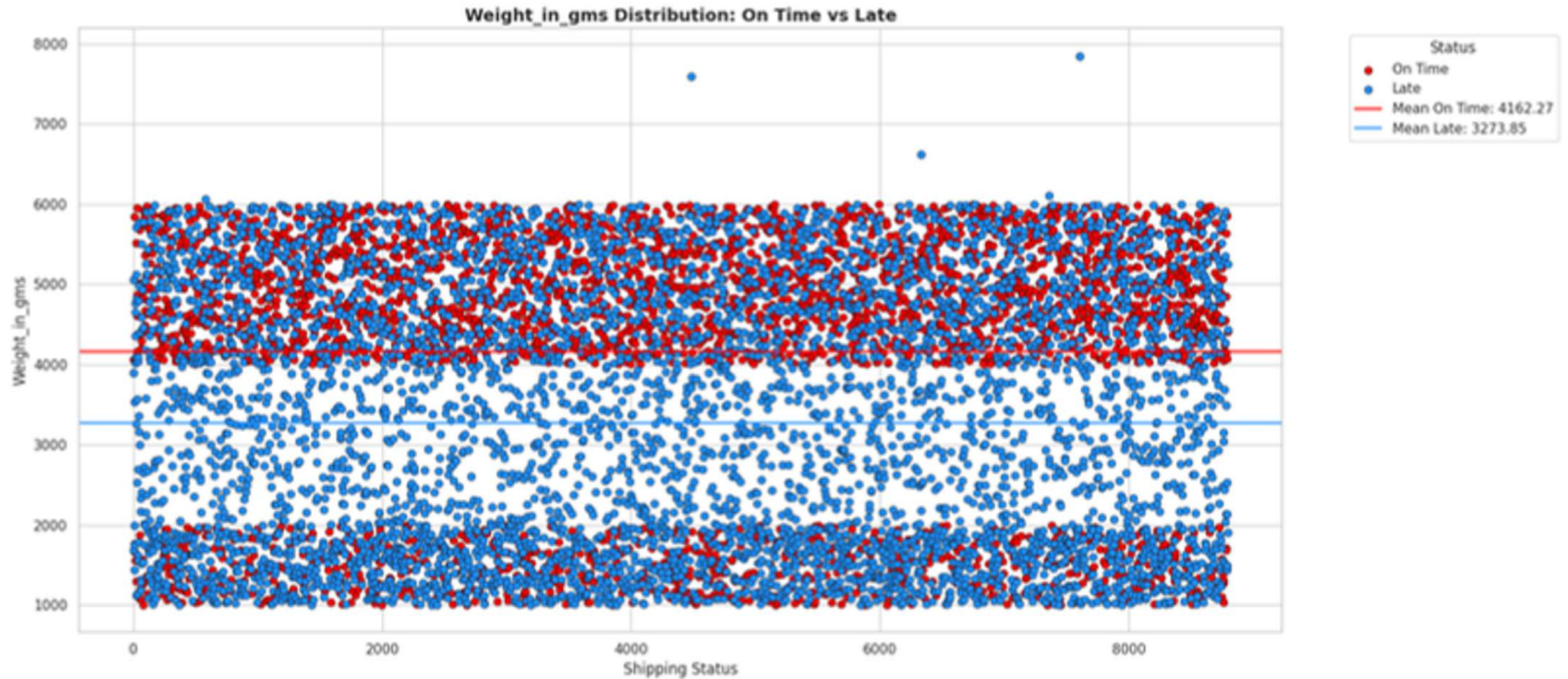
Scatter/Strip Plots

The Discount Offered Distribution: On Time vs Late



Discount_offered: Late orders average 18.5% (3x higher than on-time at 6%). Late deliveries span a wide range of 10–65%, while on-time deliveries strictly stay below 10%.

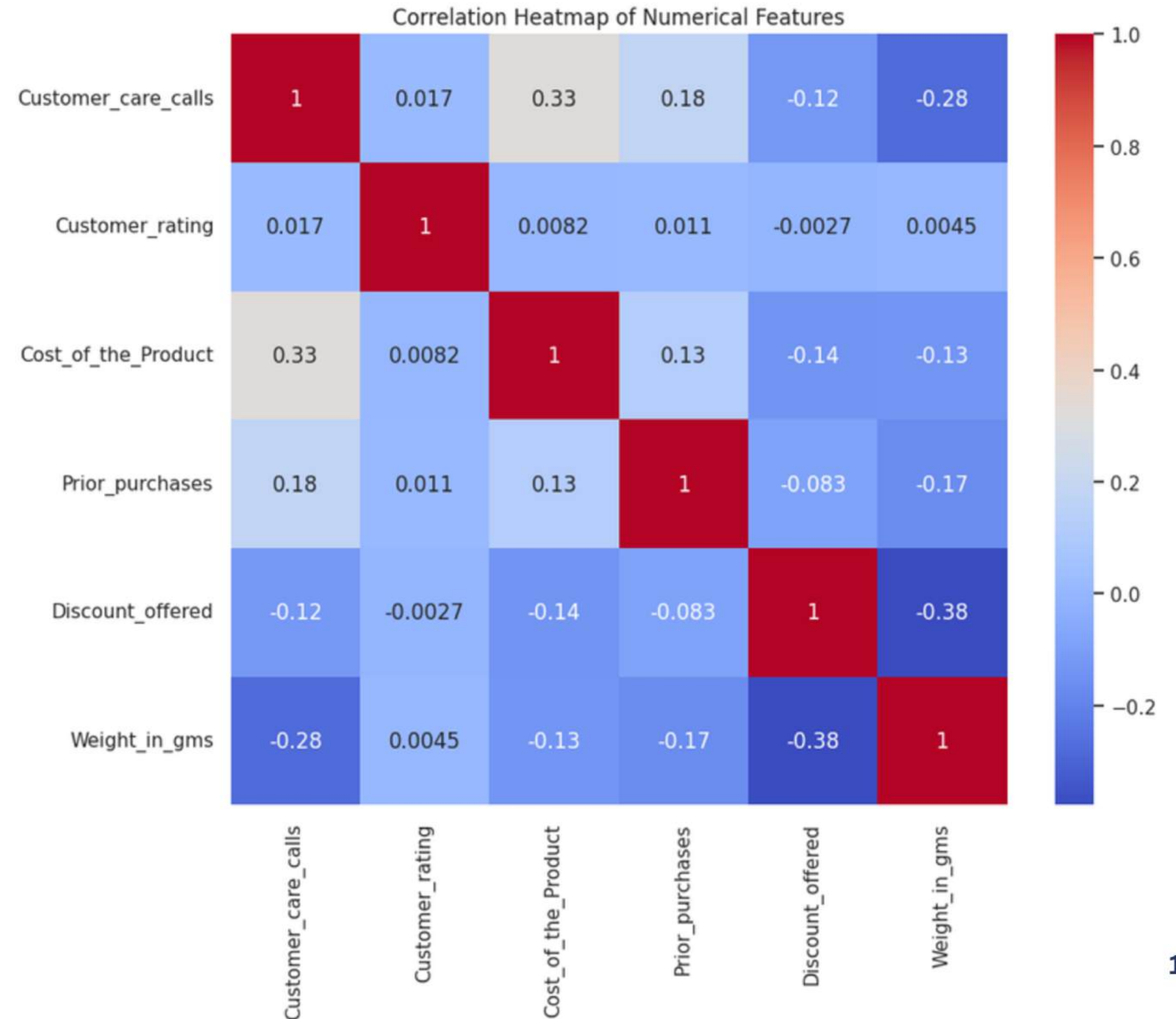
Scatter/Strip Plots



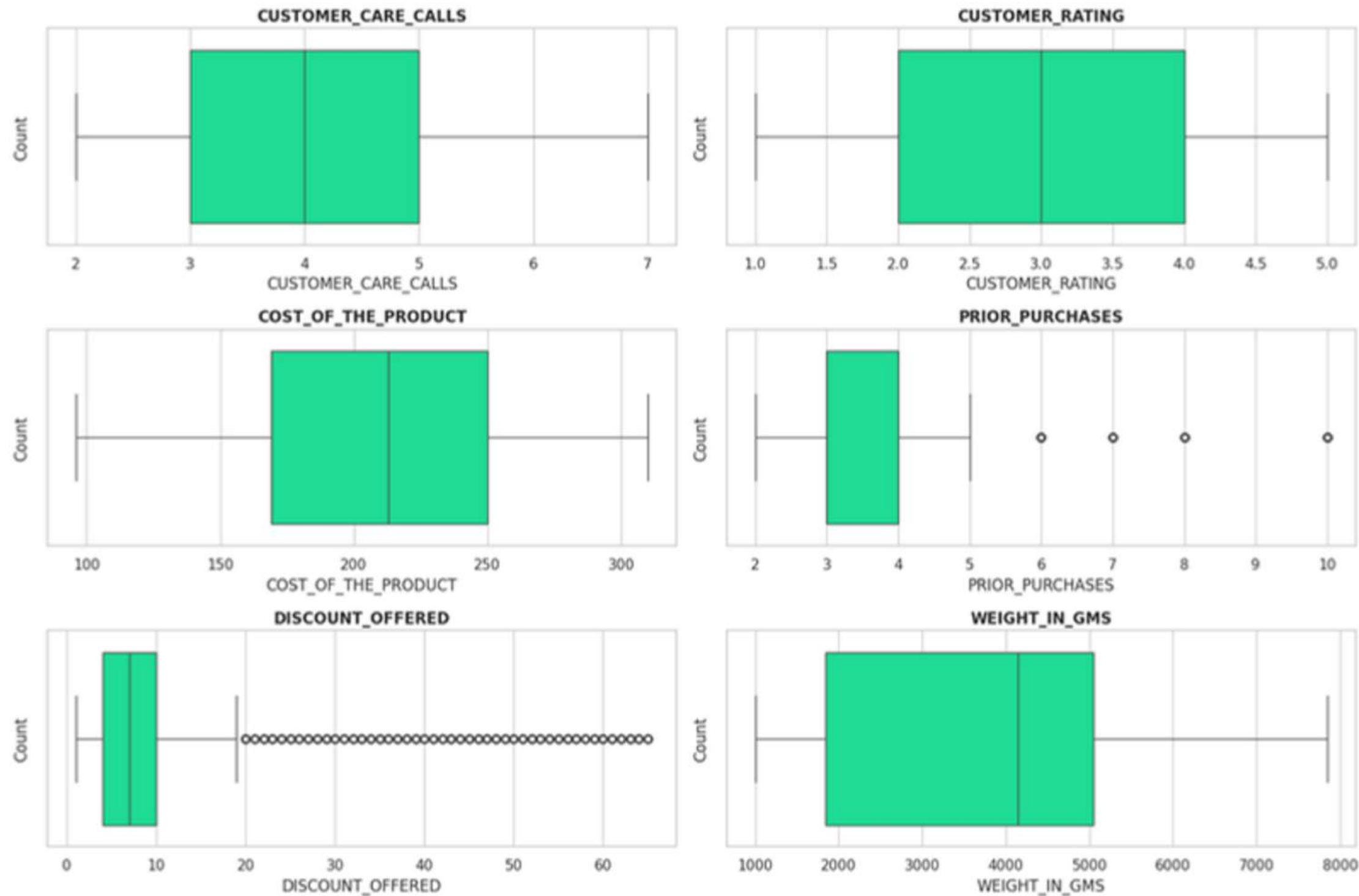
Weight_in_gms: On-time orders average significantly heavier (4,200g) compared to late orders (3,300g). This suggests heavier packages consistently achieve better on-time performance.

Correlation Heatmap

- No Serious Multicollinearity: All absolute correlations between variable pairs are < 0.8 .
- Key Target Drivers: Discount_offered and Weight_in_gms show the clearest positive correlation with the target variable.



Outlier Exploration



IQR result:

Customer_care_calls: 0 outliers
Customer_rating: 0 outliers
Cost_of_the_Product: 0 outliers
Prior_purchases: 797 outliers
Discount_offered: 1737 outliers
Weight_in_gms: 0 outliers

IQR testing identified numerous outliers in “Prior_purchases” (797) and “Discount_offered” (1,737). We decided to retain all outliers as they represent genuine business realities like wholesale buyers or flash sales rather than data errors

Data Transformation

One-Hot Encoding

We applied One-Hot Encoding to our categorical variables, carefully fitting the encoder only on the training data to prevent data leakage.

```
array(['Warehouse_block_A', 'Warehouse_block_B', 'Warehouse_block_C',  
      'Warehouse_block_D', 'Warehouse_block_F',  
      'Mode_of_Shipment_Flight', 'Mode_of_Shipment_Road',  
      'Mode_of_Shipment_Ship', 'Product_importance_high',  
      'Product_importance_low', 'Product_importance_medium'],  
      dtype=object)
```

The 3 original columns were successfully transformed into 11 new binary columns (5 Warehouse, 3 Shipment, 3 Importance).

Data Transformation

Preventing Data Leakage

Applied `.fit_transform()` strictly to the Training set to learn categories. The Test set used only `.transform()` to ensure it remained completely unseen data.

Feature Space Update

Original categorical columns were dropped after encoding to eliminate redundancy. The entire dataset is now fully digitized and model-ready.

Feature and Target Variable Separation

```
Xtrain.shape, ytrain.shape, Xtest.shape, ytest.shape
```

```
((8799, 17), (8799,), (2200, 17), (2200,))
```

Data Transformation

Standard Scaling

All features were normalized using StandardScaler (Mean=0, SD=1), fitting exclusively on the training set to strictly prevent data leakage."

ytrain, ytest

```
(0      0
1      1
2      1
3      1
4      1
..
8794    1
8795    1
8796    0
8797    1
8798    1
Name: Reached.on.Time_Y.N, Length: 8799, dtype: category
Categories (2, int64): [0, 1],
```

```
0      0
1      1
2      0
3      1
4      1
..
2195    0
2196    1
2197    1
2198    1
2199    1
Name: Reached.on.Time_Y.N, Length: 2200, dtype: category
Categories (2, int64): [0, 1])
```

Model Methodology & Selection Rationale

Methodology

- Validation Strategy: Evaluated using 10-Fold Cross-Validation on the training set.
- Primary Metric: F1-macro.

Models Tested

- Logistic Regression: Baseline linear model.
- KNN: Instance-based learning, optimized for **k**.
- SVM: Maximum margin separation using an RBF kernel.
- ANN (Chosen Model): Deep learning architecture. optimized with Adam and EarlyStopping.

```
knn = KNeighborsClassifier(n_neighbors=11)
knn_fscore = cross_val_score(knn, X_train_scaled, ytrain, scoring='f1_macro', cv=10)
```

```
▶ svc = SVC()
svc_fscore = cross_val_score(svc, X_train_scaled, ytrain, scoring='f1_macro', cv=10)
```

```
▶ lr = LogisticRegression()
lr_fscore = cross_val_score(lr, Xtrain, ytrain, scoring='f1_macro', cv=10)
```

```
model = Sequential([
    Dense(64, activation='relu', input_shape=(Xtrain.shape[1],)),
    Dropout(0.3),
    Dense(32, activation='relu'),
    Dropout(0.2),
    Dense(1, activation='sigmoid')
])

# Compile the model
model.compile(optimizer=Adam(0.001),
              loss='binary_crossentropy',
              metrics=['accuracy'])

early_stop = EarlyStopping(monitor='val_loss', patience=5, restore_best_weights=True)
```

Model Performance & Quantitative Results

Comparison Summary

- Logistic Regression: F1-macro **0.6222**, struggling with accurate single-time identification due to linear boundaries.
- KNN: F1-macro **0.6342**. But showed signs of overfitting (gap between training and CV).
- SVM: F1-macro **0.6468**. Strong traditional ML performer capturing non-linear boundaries.
- ANN: Selected as the final model for practical implementation due to the **highest** F1 training score (**0.7044**) and superior regularization.

MODEL COMPARISON (Sorted by F1-score of 'OT' class)									
Model	Train Accuracy	Test Accuracy	F1-macro	F1-OT	Recall-OT	Overfitting	Risk		
ANN (Neural Network)	0.6788	0.6741	0.6740	0.6795	0.8568		Low		
SVM	0.7002	0.6582	0.6579	0.6486	0.7824		Low		
KNN	0.7272	0.6432	0.6364	0.5866	0.6280		High		
Logistic Regression	0.6376	0.6577	0.6469	0.5851	0.5986		Low		

ANN PERFORMANCE ON TEST SET				
	precision	recall	f1-score	support
0	0.5630	0.8568	0.6795	887
1	0.8506	0.5506	0.6685	1313
accuracy			0.6741	2200
macro avg	0.7068	0.7037	0.6740	2200
weighted avg	0.7346	0.6741	0.6729	2200

Strategic Recommendations & Insights

Strengths & Limitations

- **Strength** (High Cost-Efficiency): **85.1%** precision. Generates reliable automated alerts (85 accurate per 100) without overwhelming the logistics team, successfully preventing 32.9% of total test order delays.
- **Limitation** (Coverage Volume): Misses roughly **55%** of delayed orders. Lowering the classification threshold to 0.35–0.40 could improve coverage, trading off false positives against customer dissatisfaction.

Strategic Action Plan

- **Real-time OMS Alerts:** Integrate the ANN to prioritize high-risk orders in the processing queue.
- **Logistics-Aligned Promotions:** Coordinate Marketing and Logistics before large campaigns, as Discount_offered strongly correlates with delivery delays.
- **Proactive Customer Care:** Equip customer service with tracking tools to proactively address at-risk orders, improving NPS and reducing cancellations.

Thank You for Listening

Education builds the foundation

Presented By :
Group Lazy

